

Summary: Quaternion-Based Adaptive Control for Trajectory Tracking of Quadrotor UAVs

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Contents

1	Introduction and Motivation	2
2	Equations of Motion	2
2.1	Coordinate Frames and State Variables	2
2.2	Translational Dynamics	2
2.3	Attitude Dynamics	3
2.4	Unit Quaternion Representation	3
3	Adaptive Controller Design	3
3.1	Position Adaptive Controller	3
3.2	Computation of Total Thrust and Desired Orientation	4
3.3	Attitude Adaptive Controller	5
4	Lyapunov-Based Stability Analysis	5
4.1	Closed-Loop System	5
4.2	Main Stability Theorem	6
4.3	Proof Outline	6

1 Introduction and Motivation

The paper addresses the trajectory tracking problem for a quadrotor UAV subject to **mass and inertia matrix uncertainties**. The key challenges are:

- The quadrotor is underactuated: 4 scalar inputs, 6 degrees of freedom.
- Its translational and attitude dynamics are highly nonlinear and coupled.
- Euler-angle representations suffer from kinematic singularities.
- Real systems always have parametric uncertainty.

The proposed solution is a **nonlinear hierarchical adaptive controller** that:

1. Uses the *unit quaternion* to represent orientation (singularity-free).
2. Exploits the cascade structure of the dynamics to design nested outer (position) and inner (attitude) control loops.
3. Couples the attitude controller to the translational tracking errors, allowing a unified Lyapunov stability proof without relying on cascade-system arguments.
4. Uses two Lyapunov-based adaptive laws to handle unknown mass and inertia parameters.

2 Equations of Motion

2.1 Coordinate Frames and State Variables

Let $\mathcal{I} = \{x_i, y_i, z_i\}$ be an inertial frame and $\mathcal{B} = \{x_b, y_b, z_b\}$ be the body frame fixed at the quadrotor's centre of mass. Their relative orientation is described by $\mathbf{R} \in SO(3)$, where

$$SO(3) = \{\mathbf{R} \in \mathbb{R}^{3 \times 3} \mid \mathbf{R}^T \mathbf{R} = \mathbf{I}, \det(\mathbf{R}) = 1\}.$$

The position and linear velocity in $\mathcal{I}$ are $\mathbf{p} \in \mathbb{R}^3$ and $\mathbf{v} \in \mathbb{R}^3$; the angular velocity in $\mathcal{B}$ is $\boldsymbol{\omega} \in \mathbb{R}^3$.

The skew-symmetric operator $S(\cdot) \in \mathfrak{so}(3)$ is defined for any $\boldsymbol{\vartheta} = (\vartheta_x, \vartheta_y, \vartheta_z)^T$ as

$$S(\boldsymbol{\vartheta}) = \begin{bmatrix} 0 & -\vartheta_z & \vartheta_y \\ \vartheta_z & 0 & -\vartheta_x \\ -\vartheta_y & \vartheta_x & 0 \end{bmatrix},$$

satisfying $S(\boldsymbol{\vartheta}_1)\boldsymbol{\vartheta}_2 = \boldsymbol{\vartheta}_1 \times \boldsymbol{\vartheta}_2$.

2.2 Translational Dynamics

$$\dot{\mathbf{p}} = \mathbf{v}, \tag{1}$$

$$m\dot{\mathbf{v}} = \mathcal{T} \mathbf{R} z_i - mg z_i, \tag{2}$$

where m is the (uncertain) mass, g is gravitational acceleration, $z_i = (0, 0, 1)^T$, and $\mathcal{T} = \sum_{i=1}^4 \mathcal{T}_i$ is the total thrust (scalar control input for the translational subsystem).

2.3 Attitude Dynamics

$$\dot{\mathbf{R}} = \mathbf{R} S(\boldsymbol{\omega}), \quad (3)$$

$$\mathbf{J}\dot{\boldsymbol{\omega}} = \boldsymbol{\tau} - S(\boldsymbol{\omega})\mathbf{J}\boldsymbol{\omega}, \quad (4)$$

where $\mathbf{J} \in \mathbb{R}^{3 \times 3}$ is the (uncertain) inertia matrix and $\boldsymbol{\tau} \in \mathbb{R}^3$ is the torque vector (fully actuated control input).

Property 1 (Linearity in Inertia Parameters). *The attitude dynamics are linear with respect to the inertia parameter vector $\boldsymbol{\theta} \in \mathbb{R}^p$:*

$$\mathbf{J}\dot{\boldsymbol{\omega}} + S(\boldsymbol{\omega})\mathbf{J}\boldsymbol{\omega} = \boldsymbol{\Psi}_o(\boldsymbol{\omega}, \dot{\boldsymbol{\omega}}) \boldsymbol{\theta} = \boldsymbol{\tau}, \quad (5)$$

where $\boldsymbol{\Psi}_o(\boldsymbol{\omega}, \dot{\boldsymbol{\omega}}) \in \mathbb{R}^{3 \times p}$ is the regressor matrix.

2.4 Unit Quaternion Representation

The orientation is equivalently described by the unit quaternion $\mathcal{Q} = \{\eta, \boldsymbol{\epsilon}\} \in S^3$, where $\eta \in \mathbb{R}$ is the scalar part and $\boldsymbol{\epsilon} = (\epsilon_x, \epsilon_y, \epsilon_z)^T \in \mathbb{R}^3$ is the vector part. For two quaternions $\mathcal{Q}_1 = \{\eta_1, \boldsymbol{\epsilon}_1\}$ and $\mathcal{Q}_2 = \{\eta_2, \boldsymbol{\epsilon}_2\}$, the **quaternion product** is

$$\mathcal{Q}_1 \otimes \mathcal{Q}_2 = \{\eta_1\eta_2 - \boldsymbol{\epsilon}_1^T \boldsymbol{\epsilon}_2, \eta_1\boldsymbol{\epsilon}_2 + \eta_2\boldsymbol{\epsilon}_1 + S(\boldsymbol{\epsilon}_1)\boldsymbol{\epsilon}_2\}.$$

The inverse is $\mathcal{Q}^{-1} = \{\eta, -\boldsymbol{\epsilon}\}$. The rotation matrix associated with $\mathcal{Q}$ is

$$\mathbf{R}(\eta, \boldsymbol{\epsilon}) = (\eta^2 - \boldsymbol{\epsilon}^T \boldsymbol{\epsilon})\mathbf{I} + 2\boldsymbol{\epsilon}\boldsymbol{\epsilon}^T + 2\eta S(\boldsymbol{\epsilon}). \quad (6)$$

The kinematic relationship between $\boldsymbol{\omega}$ and $\dot{\mathcal{Q}}$ is

$$\dot{\mathcal{Q}} = \frac{1}{2} \mathcal{Q} \otimes \{0, \boldsymbol{\omega}\} = \mathbf{L}(\eta, \boldsymbol{\epsilon}) \boldsymbol{\omega}, \quad \mathbf{L}(\eta, \boldsymbol{\epsilon}) \triangleq \frac{1}{2} \begin{bmatrix} -\boldsymbol{\epsilon}^T \\ \eta\mathbf{I} + S(\boldsymbol{\epsilon}) \end{bmatrix}. \quad (7)$$

3 Adaptive Controller Design

The cascade structure of the dynamics ($\mathcal{T}\mathbf{R}_{z_i}$ couples translation and attitude, while the attitude subsystem is independent of $\mathbf{p}$ and $\mathbf{v}$) permits a **hierarchical design**:

1. **Outer loop**: position adaptive controller producing a virtual force vector $\mathbf{f}$.
2. **Computation block**: derives the desired quaternion $\mathcal{Q}_d$ and total thrust $\mathcal{T}$ from $\mathbf{f}$.
3. **Inner loop**: attitude adaptive controller producing torque $\boldsymbol{\tau}$.

3.1 Position Adaptive Controller

Define the position and velocity tracking errors

$$\tilde{\mathbf{p}} = \mathbf{p}_d(t) - \mathbf{p}, \quad \tilde{\mathbf{v}} = \mathbf{v}_d(t) - \mathbf{v}. \quad (8)$$

Introduce a **reference velocity** $\mathbf{v}_r$ and **auxiliary sliding variable** $\mathbf{s}_p$:

$$\mathbf{v}_r = \mathbf{v}_d(t) + k_p \tilde{\mathbf{p}}, \quad \mathbf{s}_p = \mathbf{v}_r - \mathbf{v} = \tilde{\mathbf{v}} + k_p \tilde{\mathbf{p}}, \quad (9)$$

where $k_p > 0$.

Using the virtual input $\mathbf{f} = \mathcal{T} \mathcal{Q}_d \odot z_i = \mathcal{T} \mathbf{R}_d z_i$, the translational dynamics become

$$\dot{\tilde{\mathbf{v}}} = -k_p \tilde{\mathbf{p}} + \mathbf{s}_p, \quad (10)$$

$$m \dot{\mathbf{s}}_p = m \boldsymbol{\psi}_p(\dot{\mathbf{v}}_r) - \mathbf{f} - \mathcal{T}(\mathbf{R} - \mathbf{R}_d) z_i, \quad (11)$$

where the **translational regressor** is

$$\boldsymbol{\psi}_p(\dot{\mathbf{v}}_r) = \dot{\mathbf{v}}_r + g z_i, \quad \dot{\mathbf{v}}_r = \dot{\mathbf{v}}_d(t) + k_p \tilde{\mathbf{v}}.$$

Since mass m is unknown, the **position adaptive control law** is:

$$\boxed{\mathbf{f} = k_v \mathbf{s}_p + \alpha_p \tilde{\mathbf{p}} + \hat{m}(t) \boldsymbol{\psi}_p(\dot{\mathbf{v}}_r)}, \quad (12)$$

where $k_v, \alpha_p > 0$ are control gains and $\hat{m}(t)$ is the adaptive estimate of m , updated by:

$$\boxed{\dot{\hat{m}} = \gamma_p \boldsymbol{\psi}_p^T(\dot{\mathbf{v}}_r) \mathbf{s}_p}, \quad (13)$$

with adaptive gain $\gamma_p > 0$.

3.2 Computation of Total Thrust and Desired Orientation

The total thrust is obtained from the norm of $\mathbf{f}$:

$$\mathcal{T} = \|\mathbf{f}\|. \quad (14)$$

Define the unit vector $\mathbf{r} = \mathbf{f}/\|\mathbf{f}\| = (r_x, r_y, r_z)^T$. The desired orientation is computed by aligning the z_i -axis with $\mathbf{r}$. The corresponding quaternion is

$$\mathcal{Q}_a = \{\cos(\phi/2), \sin(\phi/2) \mathbf{a}\}, \quad (15)$$

where $\mathbf{a} = \frac{1}{\sin \phi} S(z_i) \mathbf{r} = \frac{1}{\sqrt{1-r_z^2}} (-r_y, r_x, 0)^T$ and

$$\cos(\phi/2) = \sqrt{\frac{1+r_z}{2}}, \quad \sin(\phi/2) = \sqrt{\frac{1-r_z}{2}}.$$

An additional degree of freedom (analogous to the yaw angle) $\psi_d(t)$ is incorporated via

$$\mathcal{Q}_\psi = \{\cos(\psi_d/2), \sin(\psi_d/2) z_i\}, \quad (16)$$

giving the final desired orientation quaternion:

$$\mathcal{Q}_d = \mathcal{Q}_a \otimes \mathcal{Q}_\psi. \quad (17)$$

The corresponding desired angular velocity is

$$\boldsymbol{\omega}_d(t) = \boldsymbol{\omega}_\psi(t) + \mathcal{Q}_\psi^{-1} \odot \boldsymbol{\omega}_a, \quad \boldsymbol{\omega}_\psi = (0, 0, \dot{\psi}_d)^T, \quad (18)$$

where the components of $\boldsymbol{\omega}_a$ are given by

$$\boldsymbol{\omega}_a = \frac{1}{1+r_z} \begin{bmatrix} -\dot{r}_y(1+r_z) + r_y \dot{r}_z \\ \dot{r}_x(1+r_z) - r_x \dot{r}_z \\ r_y \dot{r}_x - r_x \dot{r}_y \end{bmatrix}. \quad (19)$$

3.3 Attitude Adaptive Controller

Define the quaternion orientation error and angular velocity error:

$$\tilde{\mathcal{Q}} = \mathcal{Q}^{-1} \otimes \mathcal{Q}_d = \{\tilde{\eta}, \tilde{\epsilon}\}, \quad \tilde{\omega} = \tilde{\mathcal{Q}} \odot \omega_d(t) - \omega. \quad (20)$$

Introduce the **angular reference velocity** ω_r and **attitude sliding variable** s_o :

$$\omega_r = \tilde{\mathcal{Q}} \odot \omega_d(t) + k_\epsilon \tilde{\epsilon} + 2\alpha_o^{-1} \mathcal{T} S(\bar{\epsilon}) \mathcal{Q}^{-1} \odot s_p, \quad (21)$$

$$s_o = \omega_r - \omega = \tilde{\omega} + k_\epsilon \tilde{\epsilon} + 2\alpha_o^{-1} \mathcal{T} S(\bar{\epsilon}) \mathcal{Q}^{-1} \odot s_p, \quad (22)$$

where $k_\epsilon, \alpha_o > 0$ and $\bar{\epsilon} = (\tilde{\epsilon}_y, -\tilde{\epsilon}_x, \tilde{\eta})^T$.

The attitude error dynamics become:

$$\dot{\tilde{\mathcal{Q}}} = \mathbf{L}(\tilde{\eta}, \tilde{\epsilon}) \tilde{\omega}, \quad (23)$$

$$\dot{\tilde{\omega}} = -k_\epsilon \tilde{\epsilon} - 2\alpha_o^{-1} \mathcal{T} S(\bar{\epsilon}) \mathcal{Q}^{-1} \odot s_p + s_o, \quad (24)$$

$$\mathbf{J} \dot{s}_o = \Psi_o(\omega, \omega_r, \dot{\omega}_r) \theta - S(s_o) \mathbf{J} \omega - \tau, \quad (25)$$

where $\Psi_o(\omega, \omega_r, \dot{\omega}_r) \theta = \mathbf{J} \dot{\omega}_r + S(\omega_r) \mathbf{J} \omega$.

Since the inertia matrix $\mathbf{J}$ is unknown, the **attitude adaptive control law** is:

$$\tau = k_\omega s_o + \alpha_o \tilde{\epsilon} + \Psi_o(\omega, \omega_r, \dot{\omega}_r) \hat{\theta}(t), \quad (26)$$

with $k_\omega > 0$ and $\hat{\theta}(t) \in \mathbb{R}^p$ updated by:

$$\dot{\hat{\theta}} = \gamma_o \Psi_o^T(\omega, \omega_r, \dot{\omega}_r) s_o, \quad (27)$$

with $\gamma_o > 0$.

Remark 1. Substituting s_p and s_o back, the control laws can be written in PD-like form:

$$\mathbf{f} = k_v \tilde{\mathbf{v}} + (k_v k_p + \alpha_p) \tilde{\mathbf{p}} + \hat{m}(t) \psi_p, \quad (28)$$

$$\tau = k_\omega \tilde{\omega} + (k_\omega k_\epsilon + \alpha_o) \tilde{\epsilon} + 2k_\omega \alpha_o^{-1} \mathcal{T} S(\bar{\epsilon}) \tilde{\epsilon} \mathcal{Q}^{-1} \odot s_p + \Psi_o \hat{\theta}(t). \quad (29)$$

The gains k_v, k_p (and k_ω, k_ϵ) act as PD gains. The gains α_p, α_o shape the convergence rate. The adaptive gains γ_p, γ_o should be tuned conservatively.

4 Lyapunov-Based Stability Analysis

4.1 Closed-Loop System

Defining the mass estimation error $\tilde{m} = \hat{m}(t) - m$ and inertia estimation error $\tilde{\theta} = \hat{\theta}(t) - \theta$, the full closed-loop dynamics are:

$$\dot{\tilde{\mathbf{v}}} = -k_p \tilde{\mathbf{p}} + s_p, \quad (30)$$

$$m \dot{s}_p = -k_v s_p - \alpha_p \tilde{\mathbf{p}} - \tilde{m} \psi_p(\dot{\mathbf{v}}_r) - \mathcal{T}(\mathbf{R} - \mathbf{R}_d) z_i, \quad (31)$$

$$\dot{\tilde{m}} = \gamma_p \psi_p^T(\dot{\mathbf{v}}_p) s_p, \quad (32)$$

$$\dot{\tilde{\mathcal{Q}}} = \mathbf{L}(\tilde{\eta}, \tilde{\epsilon}) \tilde{\omega}, \quad (33)$$

$$\dot{\tilde{\omega}} = -k_\epsilon \tilde{\epsilon} - 2\alpha_o^{-1} \mathcal{T} S(\bar{\epsilon}) \mathcal{Q}^{-1} \odot s_p + s_o, \quad (34)$$

$$\mathbf{J} \dot{s}_o = -S(s_o) \mathbf{J} \omega - k_\omega s_o - \alpha_o \tilde{\epsilon} - \Psi_o(\omega, \omega_r, \dot{\omega}_r) \tilde{\theta}, \quad (35)$$

$$\dot{\tilde{\theta}} = \gamma_o \Psi_o^T(\omega, \omega_r, \dot{\omega}_r) s_o. \quad (36)$$

4.2 Main Stability Theorem

Theorem 1. Assume $\tilde{\eta}(0) \neq -1$ and consider the closed-loop dynamics (30)–(36) with adaptive laws (13) and (27). Then:

1. All closed-loop trajectories are **bounded** for all $t \geq 0$.
2. The position and attitude tracking errors converge **asymptotically to zero**:

$$\lim_{t \rightarrow \infty} \tilde{\mathbf{p}} = \mathbf{0}, \quad \lim_{t \rightarrow \infty} \tilde{\mathbf{v}} = \mathbf{0}, \quad \lim_{t \rightarrow \infty} \tilde{\boldsymbol{\epsilon}} = \mathbf{0}, \quad \lim_{t \rightarrow \infty} \tilde{\boldsymbol{\omega}} = \mathbf{0}, \quad \tilde{\eta} \rightarrow 1.$$

4.3 Proof Outline

Step 1: Interconnection term. The coupling term $(\mathbf{R} - \mathbf{R}_d)z_i$ can be expressed using the quaternion error $\tilde{\mathcal{Q}}$:

$$(\mathbf{R} - \mathbf{R}_d)z_i = \mathcal{Q} \odot (z_i - \tilde{\mathcal{Q}} \odot z_i), \quad z_i - \tilde{\mathcal{Q}} \odot z_i = 2S(\bar{\boldsymbol{\epsilon}})\tilde{\boldsymbol{\epsilon}}, \quad (37)$$

where $\bar{\boldsymbol{\epsilon}} = (\tilde{\epsilon}_y, -\tilde{\epsilon}_x, \tilde{\eta})^T$ and $\tilde{\eta}^2 + \tilde{\boldsymbol{\epsilon}}^T \tilde{\boldsymbol{\epsilon}} = 1$ has been used.

Step 2: Lyapunov function. Consider the composite Lyapunov function candidate:

$$V = \underbrace{\frac{1}{2}\alpha_p \|\tilde{\mathbf{p}}\|^2 + \frac{1}{2}m \|\mathbf{s}_p\|^2 + \frac{1}{2}\gamma_p^{-1} \tilde{m}^2}_{\text{translational part}} + \underbrace{\alpha_o \|\tilde{\boldsymbol{\epsilon}}\|^2 + \alpha_o(1 - \tilde{\eta})^2 + \frac{1}{2}\mathbf{s}_o^T \mathbf{J} \mathbf{s}_o + \frac{1}{2}\gamma_o^{-1} \|\tilde{\boldsymbol{\theta}}\|^2}_{\text{attitude part}}. \quad (38)$$

Note that $\alpha_o \|\tilde{\boldsymbol{\epsilon}}\|^2 + \alpha_o(1 - \tilde{\eta})^2$ is a valid potential on S^3 since it equals $2\alpha_o(1 - \tilde{\eta})$ (using $\tilde{\eta}^2 + \|\tilde{\boldsymbol{\epsilon}}\|^2 = 1$), which is zero only at $\tilde{\eta} = 1$.

Step 3: Time derivative of V . Computing $\dot{V}$ along (30)–(36) and using the adaptive laws to cancel the parameter uncertainty terms ($\tilde{m}$ and $\tilde{\boldsymbol{\theta}}$ cross-terms cancel exactly), one obtains:

$$\dot{V} = -\alpha_p k_p \|\tilde{\mathbf{p}}\|^2 - k_v \|\mathbf{s}_p\|^2 - 2\mathcal{T} \mathbf{s}_p^T \odot S(\bar{\boldsymbol{\epsilon}})\tilde{\boldsymbol{\epsilon}} + \alpha_o \tilde{\boldsymbol{\epsilon}}^T \tilde{\boldsymbol{\omega}} - k_\omega \|\mathbf{s}_o\|^2 - \alpha_o \mathbf{s}_o^T \tilde{\boldsymbol{\epsilon}}. \quad (39)$$

The critical cross-coupling term is cancelled using the auxiliary variable $\mathbf{s}_o$ and the identity $\tilde{\boldsymbol{\epsilon}}^T S(\bar{\boldsymbol{\epsilon}})\mathcal{Q}^{-1} \odot \mathbf{s}_p = \mathbf{s}_p^T \mathbf{R} S^T(\bar{\boldsymbol{\epsilon}})\tilde{\boldsymbol{\epsilon}} = -\mathbf{s}_p^T \odot S(\bar{\boldsymbol{\epsilon}})\tilde{\boldsymbol{\epsilon}}$, yielding:

$$\dot{V} = -\alpha_p k_p \|\tilde{\mathbf{p}}\|^2 - \alpha_o k_\epsilon \|\tilde{\boldsymbol{\epsilon}}\|^2 - k_v \|\mathbf{s}_p\|^2 - k_\omega \|\mathbf{s}_o\|^2 \leq 0. \quad (40)$$

Step 4: Boundedness. Since $\dot{V} \leq 0$, the Lyapunov function is non-increasing, which guarantees that $\tilde{\mathbf{p}}, \mathbf{s}_p, \tilde{m}, \tilde{\boldsymbol{\epsilon}}, \tilde{\eta}, \mathbf{s}_o, \tilde{\boldsymbol{\theta}}$ are all **bounded**. Consequently $\tilde{\mathbf{v}}, \mathbf{p}, \mathbf{v}, \hat{m}, \hat{\boldsymbol{\theta}}$ are also bounded. The total thrust satisfies

$$\mathcal{T} \leq \alpha_o \|\tilde{\mathbf{p}}\| + k_v \|\mathbf{s}_p\| + |\hat{m}|(\|\dot{\mathbf{v}}_d\|_\infty + k_p \|\tilde{\mathbf{v}}\| + g) < \infty, \quad (41)$$

from which it follows that $\tilde{\boldsymbol{\omega}}$ and $\boldsymbol{\omega}$ are also bounded.

Step 5: Asymptotic convergence. Define the extended state

$$\mathbf{x} = \text{col}(\tilde{\mathbf{p}}, \mathbf{s}_p, \tilde{m}, \tilde{\boldsymbol{\epsilon}}, \tilde{\eta}, \mathbf{s}_o, \tilde{\boldsymbol{\theta}}) \in \mathbb{R}^{14+p}, \quad \bar{\eta} = 1 - \tilde{\eta},$$

so that $V = \frac{1}{2}\mathbf{x}^T \mathbf{P} \mathbf{x}$ with the block-diagonal positive definite matrix

$$\mathbf{P} = \text{blockdiag}\{\mathbf{P}_p, \mathbf{P}_o\},$$

$$\mathbf{P}_p = \begin{bmatrix} \alpha_p \mathbf{I} & \mathbf{O} & 0 \\ \mathbf{O} & m \mathbf{I} & 0 \\ \mathbf{0}^T & \mathbf{0}^T & \gamma_p^{-1} \end{bmatrix}, \quad \mathbf{P}_o = \begin{bmatrix} 2\alpha_o \mathbf{I} & \mathbf{O} & \mathbf{O} & \mathbf{O} \\ \mathbf{O} & 2\alpha_o \mathbf{I} & \mathbf{O} & \mathbf{O} \\ \mathbf{O} & \mathbf{O} & \mathbf{J} & \mathbf{O} \\ \mathbf{O} & \mathbf{O} & \mathbf{O} & \gamma_o^{-1} \mathbf{I} \end{bmatrix}.$$

The function V satisfies $\frac{1}{2}\lambda_{\min}\{\mathbf{P}\}\|\mathbf{x}\|^2 \leq V(\mathbf{x}) \leq \frac{1}{2}\lambda_{\max}\{\mathbf{P}\}\|\mathbf{x}\|^2$, and $\dot{V} \leq -\mathbf{x}^T \mathbf{Q} \mathbf{x}$ where

$$\mathbf{Q} = \text{blockdiag}\{\alpha_p k_p \mathbf{I}, k_v \mathbf{I}, 0, \alpha_o k_\epsilon \mathbf{I}, 0, k_\omega \mathbf{I}, \mathbf{O}\}.$$

By Barbalat's lemma (or equivalently the auxiliary Theorem 2 in the paper, based on [?]), it follows that

$$\lim_{t \rightarrow \infty} \tilde{\mathbf{p}} = \mathbf{0}, \quad \lim_{t \rightarrow \infty} \mathbf{s}_p = \mathbf{0}, \quad \lim_{t \rightarrow \infty} \tilde{\boldsymbol{\epsilon}} = \mathbf{0}, \quad \lim_{t \rightarrow \infty} \mathbf{s}_o = \mathbf{0}.$$

This in turn implies $\tilde{\mathbf{v}} \rightarrow \mathbf{0}$ and $\tilde{\boldsymbol{\omega}} \rightarrow \mathbf{0}$. Since $\tilde{\eta}^2 + \|\tilde{\boldsymbol{\epsilon}}\|^2 = 1$ and $\tilde{\eta}(0) \neq -1$, the scalar part satisfies $\tilde{\eta} \rightarrow +1$ (the stable equilibrium), completing the proof.